

Fertility and Housing Tenure Choice: Quantitative One-Market Model Writeup

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1 Model

This version rewrites the May 2026 model as a non-spatial quantitative lifecycle economy. The May draft had residential locations, local amenities, bilateral moving wedges, and location-specific housing supply. The current model removes that spatial layer. There is one housing-services market, one purchase price, and one rental user cost. What remains is the channel that is central for the quantitative exercise: children require goods and housing space, family-suitable housing is easier to access through ownership than through renting, and the ownership margin is shaped by down-payment constraints, transaction costs, bequest motives, and tax incentives to retain housing.

The object is a stationary overlapping-generations economy. Adult population mass is normalized in the benchmark, so fertility changes the composition of households and their housing demand rather than mechanically changing aggregate city scale. This normalization keeps the quantitative model focused on the family-housing margin. A separate population-scale closure can be added later, but it is not needed to define the current equilibrium.

The model combines endogenous fertility in a lifecycle problem (Sommer, 2016; Doepke and Kindermann, 2019) with housing tenure choice, collateral constraints, and owner-occupied housing (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). Relative to the May draft, the Rosen–Roback spatial layer is removed: there are no local amenities, no bilateral moving wedges, and no location-specific housing markets in the baseline quantitative economy.

1.1 Environment

Agents. Households live for ages $a = 1, \dots, J$. They work until age J_R and are retired afterward. At the beginning of a period, the core individual state is

$$x = (b, z, d, a, n, s). \tag{1}$$

Here b is liquid wealth, $z \in \mathcal{Z}$ is idiosyncratic labor productivity, $d \in \mathcal{D}$ is tenure and housing position, a is age, n is completed fertility, and $s \in \mathcal{S}$ is the child-at-home state. The tenure set is

$$\mathcal{D} = \{R, O_1, \dots, O_K\}.$$

R denotes renting. O_k denotes ownership of a house with physical size H_k , where $H_1 < \dots < H_K$. Renters choose rental space continuously up to a cap $\bar{h}^R$, while owners choose from the discrete

ladder $\{H_k\}_{k=1}^K$. The ladder is the reduced-form way the quantitative model captures the fact that large family-suitable units are disproportionately owner-occupied.

When the capital-gains module is active, the owner state is augmented by a tax basis B^p . The baseline implementation can be recovered by setting the capital-gains tax rate to zero, in which case B^p drops out of the household problem.

Income. Working-age households receive after-tax labor income

$$y_a(z) = (1 - \tau^{\text{pay}})w e_a z, \quad a \leq J_R,$$

where e_a is the deterministic lifecycle efficiency profile and z follows the Markov transition matrix $\Pi^z(z'|z)$. Retired households receive pension income p_a^{ret} . The quantitative model treats the income process as a calibrated household primitive rather than as the source of a spatial wage gradient.

Preferences. Let $m(n, s)$ be the number of dependent children currently living at home. Completed fertility n and current dependents $m(n, s)$ differ because children eventually leave the household. Child-related goods and housing needs are

$$\begin{aligned} \bar{c}(n, s) &= \bar{c}_0 + \bar{c}_n m(n, s), \\ \bar{h}(n, s) &= \bar{h}_0 + \mathbb{I}\{m(n, s) \geq 1\} \bar{h}_{\text{jump}} + \bar{h}_n m(n, s). \end{aligned} \tag{2}$$

The first child can therefore create a discrete housing-space requirement, and additional children raise required space linearly.

The flow utility aggregator is Stone–Geary over nondurable consumption and housing services:

$$u(c, \tilde{h}; n, s) = \frac{\left[(c - \bar{c}(n, s))^\alpha \tilde{h}^{1-\alpha} \right]^{1-\sigma}}{1 - \sigma} \psi_{\text{child}} m(n, s), \tag{3}$$

with utility equal to $-\infty$ if consumption or housing services fall below the required levels. The object $\tilde{h}$ is residual housing service after child space needs. For renters and owners,

$$\tilde{h} = \begin{cases} h^R - \bar{h}(n, s), & d = R, \\ \chi [H_k - \bar{h}(n, s)], & d = O_k. \end{cases} \tag{4}$$

The parameter χ is an owner-occupancy service premium applied to residual owner housing services. In market clearing, owner demand is the physical quantity H_k , not the service quantity χH_k .

The key fertility mechanism is already visible in (3) and (4). Children do not only require goods. They also absorb scarce housing space. If a household is constrained by the rental cap, the ownership menu, or the down-payment requirement, the relevant price of a child is the household's shadow value of space, not only the market rent.

Bequests. At death, households receive warm-glow bequest utility. Let W^B denote bequeathable wealth after any liquidation costs and taxes. Bequest utility is

$$\mathcal{B}(W^B, n) = \theta_0 (1 + \theta_n n) \frac{(\theta_1 + \max\{W^B, 0\})^{1-\sigma} - \theta_1^{1-\sigma}}{1 - \sigma}. \tag{5}$$

The parameter θ_0 governs the overall bequest motive, θ_n lets completed fertility scale the bequest motive, and θ_1 controls curvature near zero wealth.

Housing market. There is one aggregate housing-services market. The rental user cost is q and the asset price of housing is P . They satisfy

$$q = (r + \delta + \tau^p)P, \quad (6)$$

where r is the risk-free interest rate, δ is maintenance/depreciation, and τ^p is the recurrent property-tax rate. Aggregate housing supply is upward sloping:

$$H^S(q) = H_0 \left(\frac{q}{\bar{q}} \right)^\eta. \quad (7)$$

The scale H_0 fixes the benchmark level of housing services and η governs the construction response.

Financing and tenure institutions. Liquid wealth earns gross return $R = 1 + r$. Renters cannot borrow. A household that originates an owner position O_k can finance at most share ϕ of the house value. Equivalently, the required down payment is

$$(1 - \phi)PH_k. \quad (8)$$

The parameter ϕ is the financed share, so a larger ϕ relaxes the ownership constraint.

Selling a house is costly. If a household sells a house of size H_k , sale proceeds before capital-gains taxes are

$$(1 - \psi^s)PH_k,$$

where ψ^s is the proportional sale or transaction wedge. A household that keeps its current house is not forced to satisfy a new origination constraint after a price change. This distinction matters for old owners: the model can generate retention because selling is costly, because current owners already hold large units, and because tax rules can make holding more attractive than realizing gains.

Capital gains and step-up basis. The capital-gains block is modular. Let B^p be the tax basis attached to an owner-occupied house and let $\bar{g}$ be the exempt gain. If the house is sold during life, the tax liability is

$$\mathcal{T}^g(PH_k, B^p) = \tau^g \max\{PH_k - B^p - \bar{g}, 0\}. \quad (9)$$

Sale proceeds net of transaction costs and capital-gains taxes are therefore

$$\Omega(H_k, B^p) = (1 - \psi^s)PH_k - \mathcal{T}^g(PH_k, B^p). \quad (10)$$

Under step-up basis, accrued gains are not taxed at death. The tax basis for heirs is reset to the market value of the inherited house. This creates a retention incentive for older owners with large unrealized gains: selling converts the gain into a current tax liability, while holding until death can forgive that liability. Setting $\tau^g = 0$ or $B^p = PH_k$ collapses this block to the current no-capital-gains baseline.

1.2 Household Problem

Within a period, the household first faces the fertility choice if eligible, then chooses tenure and housing position, and finally chooses consumption, rental space, and saving. The order is useful

because fertility changes housing needs before the household chooses whether to rent, keep an existing home, or buy a different owner rung.

Housing adjustment. Let $H(R) = 0$ and $H(O_k) = H_k$. Given current tenure d and next tenure d' , post-adjustment liquid wealth is

$$\tilde{b}(b, d, d') = b + \mathbb{I}\{d \in \mathcal{O}, d' \neq d\} \Omega(H(d), B^p) - \mathbb{I}\{d' \in \mathcal{O}, d' \neq d\} PH(d'), \quad (11)$$

where $\mathcal{O} = \{O_1, \dots, O_K\}$. If the capital-gains block is inactive, the argument B^p is suppressed and $\Omega(H(d), B^p) = (1 - \psi^s)PH(d)$.

The feasible next-tenure set is

$$\mathcal{D}(b, d) = \left\{ d' \in \mathcal{D} : \tilde{b}(b, d, d') \geq \underline{b}(d'|b, d) \right\}. \quad (12)$$

For renters and new owner positions,

$$\underline{b}(R|b, d) = 0, \quad \underline{b}(O_k|b, d) = -\phi PH_k.$$

For an incumbent owner who keeps the same house, the lower bound is

$$\underline{b}(O_k|b, O_k) = \min\{b, -\phi PH_k\}. \quad (13)$$

Thus existing owners are not forced to delever mechanically when prices move, but households that buy or upsize must satisfy the origination down-payment constraint.

Branch values. Let V_{a+1} be next period's value. The continuation value after income risk and child aging is

$$\bar{V}_{a+1}(b', d', z, n, s) = \sum_{z' \in \mathcal{Z}} \sum_{s' \in \mathcal{S}} \Pi^z(z'|z) \Pi^c(s'|s, n) V_{a+1}(b', z', d', a+1, n, s'). \quad (14)$$

At the terminal date, $V_{J+1}(b, z, d, J+1, n, s) = \mathcal{B}(W^B, n)$.

Conditional on choosing to rent, the household solves

$$\begin{aligned} V_a^R(\tilde{b}, z, n, s) &= \max_{c, h^R, b'} u(c, h^R - \bar{h}(n, s); n, s) + \beta \bar{V}_{a+1}(b', R, z, n, s) \\ \text{s.t.} \quad &c + qh^R + b' = R\tilde{b} + y_a(z), \\ &0 \leq h^R \leq \bar{h}^R, \quad b' \geq 0. \end{aligned} \quad (15)$$

The rental cap is a primitive family-housing access margin: when it binds, the household's marginal value of space can exceed the market rent.

Conditional on owning rung O_k , the household solves

$$\begin{aligned} V_a^{O_k}(\tilde{b}, z, n, s) &= \max_{c, b'} u(c, \chi[H_k - \bar{h}(n, s)]; n, s) + \beta \bar{V}_{a+1}(b', O_k, z, n, s) \\ \text{s.t.} \quad &c + (\delta + \tau^p)PH_k + b' = R\tilde{b} + y_a(z), \\ &b' \geq \underline{b}^+(O_k). \end{aligned} \quad (16)$$

The lower bound $\underline{b}^+(O_k)$ is the continuation borrowing limit associated with the owner position. The owner pays recurrent property taxes and maintenance through the service cost $(\delta + \tau^p)PH_k$.

Tenure and house-size choice. For each feasible next tenure d' , define the deterministic branch value

$$\tilde{V}_a^T(d'|b, z, d, n, s) = \begin{cases} V_a^R(\tilde{b}(b, d, R), z, n, s), & d' = R, \\ V_a^{O_k}(\tilde{b}(b, d, O_k), z, n, s), & d' = O_k. \end{cases} \quad (17)$$

Tenure and owner-rung choices are smoothed with Type-I extreme-value shocks of scale κ_T . The probability of choosing d' is

$$\pi_a^T(d'|b, z, d, n, s) = \frac{\exp\left(\tilde{V}_a^T(d'|b, z, d, n, s)/\kappa_T\right)}{\sum_{\hat{d} \in \mathcal{D}(b, d)} \exp\left(\tilde{V}_a^T(\hat{d}|b, z, d, n, s)/\kappa_T\right)}, \quad (18)$$

and the inclusive tenure value is

$$V_a^T(b, z, d, n, s) = \kappa_T \log \sum_{\hat{d} \in \mathcal{D}(b, d)} \exp\left(\tilde{V}_a^T(\hat{d}|b, z, d, n, s)/\kappa_T\right). \quad (19)$$

As κ_T approaches zero, tenure choice converges to the deterministic maximum. With positive κ_T , the model allows small residual dispersion in tenure and size choices.

Fertility. During the fertile window

$$a \in \{A_f^{\text{start}}, \dots, A_f^{\text{end}}\},$$

a childless household chooses $n' \in \mathcal{N} = \{0, 1, \dots, \bar{n}\}$. The option $n' = 0$ means no birth in the current period. A positive option fixes completed fertility forever and moves the household into the first dependent-child state. Let

$$s(n') = \begin{cases} 0, & n' = 0, \\ 1, & n' > 0. \end{cases}$$

The value of fertility option n' is

$$\tilde{V}_a^F(n'|b, z, d) = V_a^T(b, z, d, n', s(n')). \quad (20)$$

Fertility choices are smoothed with Type-I extreme-value shocks of scale κ_F :

$$\pi_a^F(n'|b, z, d) = \frac{\exp\left(\tilde{V}_a^F(n'|b, z, d)/\kappa_F\right)}{\sum_{\hat{n} \in \mathcal{N}} \exp\left(\tilde{V}_a^F(\hat{n}|b, z, d)/\kappa_F\right)}. \quad (21)$$

The household value at a childless fertile state is

$$V_a(b, z, d, a, 0, 0) = \kappa_F \log \sum_{\hat{n} \in \mathcal{N}} \exp\left(\tilde{V}_a^F(\hat{n}|b, z, d)/\kappa_F\right). \quad (22)$$

Outside the fertile window, and after a positive fertility choice has already been made, fertility is degenerate and $V_a(b, z, d, a, n, s) = V_a^T(b, z, d, n, s)$.

This block should be interpreted as a completed-family choice in the current quantitative model. It is not a sequential second-birth hazard model. The sequential parity version is a natural extension, but it requires a larger fertility state space and different empirical moments.

1.3 Stationary Equilibrium

The equilibrium is stationary in prices, household policies, and the normalized distribution of adult households. The aggregate adult mass is normalized to one. This normalization means the model clears the housing market for a stationary composition, not for an endogenous city size.

Definition 1 (Stationary equilibrium). *A stationary equilibrium is a collection*

$$\mathcal{X}^* = (P, q, \{V_a, V_a^R, V_a^{O_k}, V_a^T, \tilde{V}_a^T, \tilde{V}_a^F\}_{a,k}, \{c, h^R, b', d', n'\}, \{\pi^T, \pi^F\}, g, H^D, H^S)$$

such that:

1. **Household optimization.** *Given prices, taxes, and the income process, value functions solve the household problem in (15)–(16), with tenure probabilities (18), fertility probabilities (21), and terminal bequest utility (5). Policies for consumption, rental space, saving, tenure, owner rung, and fertility are associated optimal policies.*
2. **Stationary distribution.** *The normalized adult distribution g is invariant under the transition induced by optimal policies, tenure and fertility probabilities, the income transition, child aging, deterministic aging, and death. Adult mass is normalized:*

$$\int dg(x) = 1.$$

New adult households enter at the initial age with the benchmark distribution over income, assets, tenure, fertility, and child state.

3. **Housing demand.** *Aggregate physical housing demand is*

$$H^D = \int \left[\mathbb{I}\{d = R\} h^R(x) + \sum_{k=1}^K \mathbb{I}\{d = O_k\} H_k \right] dg(x). \quad (23)$$

Housing demand counts physical space. It does not multiply owner-occupied units by the service premium χ .

4. **Housing market clearing and user cost.** *The aggregate housing market clears:*

$$H^D = H^S(q), \quad q = (r + \delta + \tau^P)P. \quad (24)$$

5. **Government accounting.** *Payroll taxes finance the pension profile if the pension block is closed inside the model. Property taxes and capital-gains taxes are rebated or absorbed by the government according to the counterfactual experiment. When the capital-gains module is inactive, $\tau^g = 0$ and the tax basis state is irrelevant.*

2 Quantitative Content

The quantitative model is designed to connect housing access to fertility through a small number of observable margins. The calibration should discipline the object that appears in the mechanism: the shadow price of family-suitable housing faced by young households.

2.1 Moment Blocks

The paper-facing calibration table should be organized around parameter blocks, not around the order in which the code computes moments:

1. **Fertility.** Completed fertility, total fertility, and childlessness discipline the net value of children, the curvature of the fertility choice, and the child cost parameters.
2. **Lifecycle tenure.** Ownership by age and old-age ownership discipline down-payment constraints, transaction costs, and the bequest or retention motives.
3. **Family housing demand.** Housing-size changes around children, rooms or housing services by child status, and the renter-owner size gap discipline $\bar{h}_{\text{jump}}$, $\bar{h}_n$, the renter cap $\bar{h}^R$, and the owner size ladder.
4. **Market clearing and supply.** The level of housing services and the user cost identify the supply scale H_0 and the price response governed by η .
5. **Wealth and bequests.** Assets by age, housing wealth at old ages, and bequest-related moments discipline the warm-glow bequest parameters and the strength of retention at the end of life.

This organization keeps the calibration tied to the economics. Fertility moments alone do not identify the housing block. Housing-size moments alone do not identify the value of children. Old-age ownership is informative about retention only once the transaction, tax, and bequest channels are stated explicitly.

2.2 Counterfactuals

The natural counterfactuals change one primitive access margin at a time:

1. relax the down-payment constraint by increasing ϕ ;
2. expand rental access to family-sized housing by increasing $\bar{h}^R$;
3. reduce transaction or sale wedges ψ^s ;
4. remove or change capital-gains taxation and step-up basis;
5. increase housing supply elasticity η or the supply scale H_0 ;
6. change child-related housing needs $\bar{h}_{\text{jump}}$ and $\bar{h}_n$ to measure the importance of the space requirement itself.

These experiments should be interpreted as general-equilibrium exercises in the one housing market. They change prices through (24), but they do not include migration, spatial sorting, or city-scale feedbacks.

3 Relation to the May Spatial Draft

The May model had two locations, local amenities, local housing supply, and an outside-option entry system for city size. Those objects are deliberately absent from the current quantitative model. The mechanism is now national or one-market rather than spatial: the relevant scarcity is access to family-sized housing services, especially through the tenure system, not the allocation of families across center and periphery.

The substantive elements retained from the May draft are the lifecycle problem, the renter-owner distinction, the owner size ladder, the rental-size cap, the down-payment constraint, child housing needs, bequest motives, and housing market clearing. The main new element relative to the May presentation is the capital-gains and step-up-basis module. That module is useful because it creates a direct old-owner retention wedge: an older household with large unrealized gains may hold a large house even when its current housing need is low, because selling triggers taxes that can be avoided by holding until death.

The current draft should therefore not claim that old occupancy is inefficient by itself. A large house occupied by an old owner is inefficient only relative to a stated friction or policy wedge: transaction costs, capital-gains taxes, step-up basis, property-tax preferences, or a missing market that prevents young families from accessing equivalent space. The quantitative task is to measure how much those wedges raise young households' shadow value of family housing and how much that shadow value depresses fertility.

A Optional Population Closure

The benchmark normalization fixes adult mass at one. If the paper later needs a scale channel, the least disruptive extension is a renewal equation for young adult entry. Let E_0 be the mass of new adult entrants required to reproduce one unit of the stationary age distribution, and let B_0 be the mature-child flow produced by one unit of adult mass. With outside-born potential entrant flow M and retention probability ρ , stationary scale would satisfy

$$SE_0 = M + \rho SB_0, \quad S = \frac{M}{E_0 - \rho B_0}. \quad (25)$$

This adds a scale response to fertility without reintroducing locations. It is not part of the baseline quantitative model because the current calibration targets the composition of households and housing choices, not city population growth.

References

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